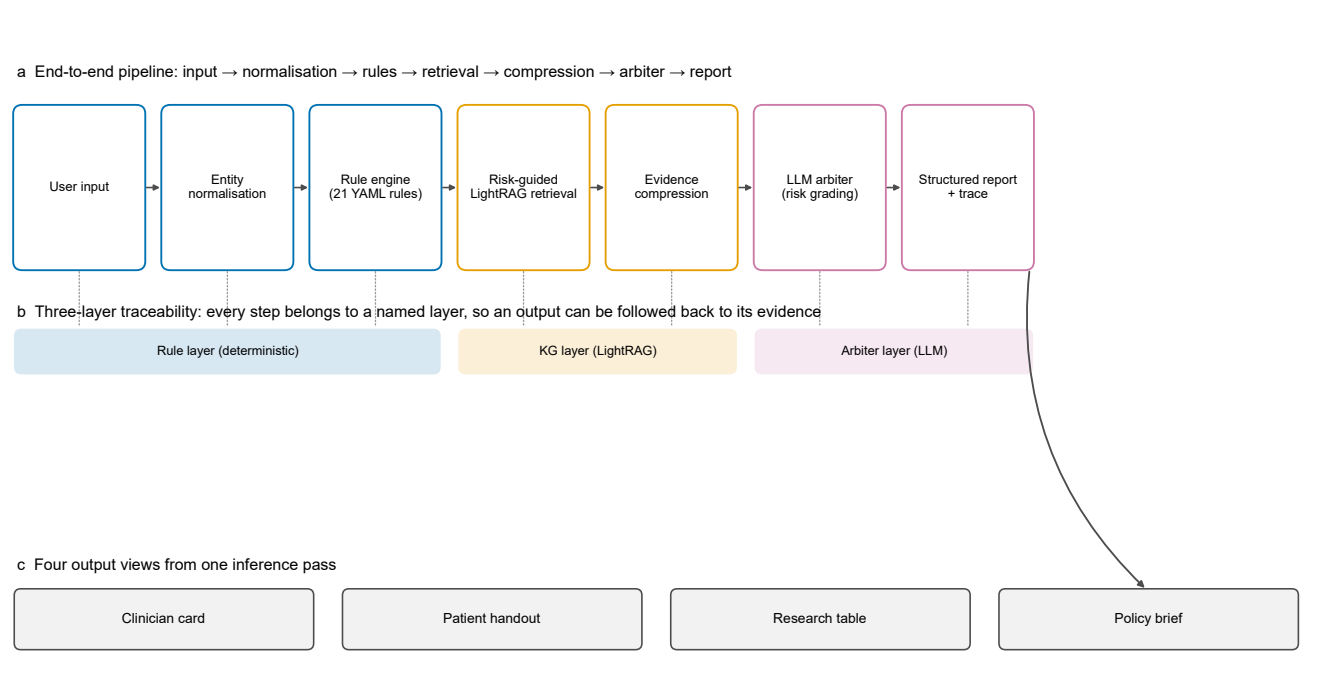
a End-to-end pipeline: input → normalisation → rules → retrieval → compression → arbiter → report
User input
Entity normalisation
Rule engine (21 YAML rules)
Risk-guided LightRAG retrieval
Evidence compression
LLM arbiter (risk grading)
Structured report + trace
b Three-layer traceability: every step belongs to a named layer, so an output can be followed back to its evidence
Rule layer (deterministic)
KG layer (LightRAG)
Arbiter layer (LLM)
c Four output views from one inference pass
Clinician card
Patient handout
Research table
Policy brief